C Language — Investigative Question Bank

This document contains elaborated C programming questions only. There are no solutions or explanations. Each question is designed to force reasoning, experimentation, and debugging.

Star Pattern Problems (10)

1. Print a hollow diamond star pattern with user-defined height using only nested loops.

2. Generate a butterfly star pattern that expands and contracts symmetrically.

3. Print a right-aligned Pascal-style star triangle without extra storage.

4. Create an hourglass star pattern using minimal loops.

5. Print a square star pattern with diagonals crossing at the center.

6. Generate concentric square star patterns based on input layers.

7. Print a zig-zag star pattern across three rows without modulo operator.

8. Create an X-shaped star pattern inside a square grid of odd size.

9. Print stars only at prime positions in each row.

10. Design a star pattern where total stars printed are minimized.

Pointer Problems (20)

11. Traverse an array using pointers only, no indexing.

12. Demonstrate pointer and pointer-to-pointer differences.

13. Swap two integers using double pointers.

14. Reverse an array using pointer arithmetic.

15. Return a pointer to the maximum element in an array.

16. Find string length using pointers only.

17. Explain sizeof array vs sizeof pointer using code.

18. Modify a variable using its address in a function.

19. Demonstrate pointer arithmetic on different data types.

20. Compare two pointers and explain validity.

21. Use void pointers to store multiple data types.

22. Create a generic swap using void pointers.

23. Create and observe dangling pointers.

24. Demonstrate use-after-free.

25. Show pointer aliasing effects.

26. Allocate memory dynamically and access via pointers.

27. Return pointer to an array from a function.

28. Demonstrate pointer decay in functions.

29. Print memory addresses in hexadecimal.

30. Differentiate NULL pointer vs uninitialized pointer.

Array Problems (20)

31. Rotate an array left by k positions in-place.

32. Find second largest and smallest in one traversal.

33. Merge two sorted arrays.

34. Remove duplicates preserving order.

35. Count frequency of elements using arrays.

36. Reverse an array using recursion.

37. Check if array is palindrome.

38. Separate even and odd elements maintaining order.

39. Find missing elements in a sequence.

40. Alternate positive and negative numbers.

41. Find equilibrium index of array.

42. Compute prefix sums.

43. Find common elements of two arrays.

44. Move zeros to end maintaining order.

45. Find longest increasing contiguous subarray.

46. Rotate 2D matrix by 90 degrees.

47. Find row with maximum 1s in binary matrix.

48. Check if two arrays are equal ignoring order.

49. Implement manual array resizing.

50. Detect duplicate elements.

Pointers + Arrays (10)

51. Traverse a 2D array using pointers only.

52. Modify 2D array elements using pointer arithmetic.

53. Create dynamic array using malloc and pointers.

54. Modify array via function using pointers.

55. Sum array elements using pointer traversal.

56. Binary search using pointer arithmetic.

57. Reverse string using pointers.

58. Find max element in 2D array using pointers.

59. Demonstrate pointer decay in multidimensional arrays.

60. Swap rows of a matrix using pointers.

Nested Loop Problems (10)

61. Print multiplication table using nested loops.

62. Generate Floyd’s triangle.

63. Print prime numbers in a range.

64. Create number pyramid pattern.

65. Print cyclic shift matrix.

66. Find all pairs with given sum.

67. Create increasing-decreasing number pattern.

68. Generate hollow square number pattern.

69. Print all combinations of two numbers.

70. Generate cumulative sum pattern.

Heap / Dynamic Memory Problems (10)

71. Allocate memory dynamically and compute sum.

At Last

72. Resize dynamic array using realloc.
73. Demonstrate memory leak.
74. Free memory conditionally and analyze safety.
75. Reverse dynamically allocated string.
76. Compare malloc vs calloc initialization.
77. Simulate low-memory conditions.
78. Demonstrate double free error.
79. Create safe malloc wrapper.
80. Track allocated and freed memory.

Function Problems (10)

81. Factorial using recursion.

82. Return multiple values using pointers.

83. Menu-driven program using functions.

84. Call by reference modification.

85. Use static variable inside function.

86. Reverse number using recursion.

87. Process array inside function.

88. Demonstrate call stack using recursion.

89. Validate user input using function.

90. Use function pointers.

Structure Problems (10)

91. Student structure with multiple records.

92. Array of structures with formatted output.

93. Pass structure by value and reference.

94. Demonstrate structure padding.

95. Compare structure and union memory.

96. Nested structures.

97. Dynamic allocation of structure.

98. Use typedef with structures.

99. Sort structure records.

100. File I/O with structures.

Multi-File Linking Problems (10)

101. Split program into two source files.

102. Share variable using extern.

103. Effect of static on global variables.

104. Create and use header file.

105. Fix multiple definition errors.

106. Fix undefined reference errors.

107. Organize project with headers.

108. Create reusable module.

109. Compile multiple files using gcc.

110. Create Makefile for project.